ECON 0100 | Exercise D3 | Public Goods

Solutions

Answers and work are shown in the red blocks.

Q1 | The Dumbledore Memorial

Three Dumbledores were considering whether to install a memorial of their brother Albus just outside their town. The individual marginal benefits for three ~~Dumbledors~~{++Dumbledores++} is below.

Aurelius	Aberforth	Ariana
75	10	20

Since the statue will be installed in public space, any of the three are able to enjoy the statue regardless of whether they helped pay for it. It costs 90 coins per year to operate and maintain the memorial.

a) Visit Count

How many of the three Dumbledores would visit the memorial if it were built?

All three have a positive marginal benefit for visiting. So all three will visit.

b) Continued Funding

There's an endowment to fund the memorial for ~~it's~~{++its++} first year. Simply considering their financial interests, how many Dumbledores will contribute to the memorial's continued operation after the endowment runs out?

None. This is a classic free rider problem.

c) The Referendum

Aurelius proposes a referendum that would raise residents' taxes by 30 coins per year, and puts it to a vote. If each ~~Dumbledor~~{++Dumbledore++} voted in their own private interests, would this referendum receive a majority vote?

No. Only Aurelius would vote for the referendum. So it will not pass.

d) Marginal Social Benefit

What is the marginal social benefit from the memorial?

$MSB = 75 + 10 + 20 = 105$

e) Efficiency

If the referendum passed, would it lead to an efficient outcome?

Yes. The MSB is greater than the $MSC = 90$. So unlike with Dobby's Memorial, this one will not be built even though it is socially efficient. Referenda don't always implement the social planner.

Q2 | A Memorial to Dobby

Hogsmeade residents were considering whether to install a memorial of Dobby, the heroic house elf, in their town. The individual marginal benefits for five recently graduated residents is below.

Harry	Ron	Hermione	Ginny	Neville
34	29	24	19	14

Any of the five are able to enjoy the statue, regardless of whether they helped pay for it. And it costs 100 coins per year to operate and maintain the memorial.

a) Visit Count

How many of these five residents would visit the memorial if it were built?

All five will visit: each has a positive marginal benefit, and no one can be kept out.

b) The Referendum

Neville proposes a referendum that would raise residents' taxes by 20 coins per year, and puts it to a vote. If each of the five Hogsmeade citizens voted in their own private interests, would this referendum receive a majority vote?

Yes! Harry, Ron and Hermione each value the memorial above the 20 coin tax: 3 to 2.

c) Marginal Social Benefit

What is the marginal social benefit from the memorial?

$MSB = 34 + 29 + 24 + 19 + 14 = 120$

d) Efficiency

Would this referendum lead to an efficient outcome?

Yes. $MSB = 120 > MSC = 100$.

e) A Higher Tax

Would the proposal receive a majority vote if the tax were 25 coins?

No. Hermione would vote no, and only Harry and Ron value it above 25: 2 to 3. It fails even though the memorial is still efficient.

f) Admission

Hermione proposes allowing a private company to build the memorial and charge residents a lump-sum fee to visit the memorial as they like. If the private company were allowed to set a single fee, would the company be able to cover its costs?

No. There's no price that would cover the costs, that's high enough to cover costs and low enough so that enough residents would participate: a fee of 20 collects $3 \times 20 = 60$, a fee of 24 collects $3 \times 24 = 72$, and a fee of 19 collects $4 \times 19 = 76$, all short of 100.